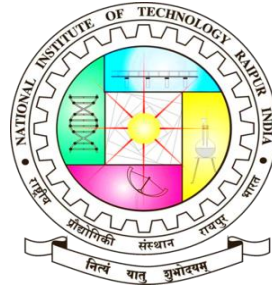


Major Project Presentation on **Application of Ensemble Learning Model for Disease Prediction**



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Outline

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INTRODUCTION:

- Disease diagnosis refers to the process of determining which disease best reflects a person's symptoms. The most challenging issue is diagnosis because certain symptoms and indications are vague, and disease identification is vital in treating any sickness.
- In the modern era, Major Chronic Disease (MCD), including heart disease, diabetes, kidney disease, etc., is responsible for nearly 89% of deaths in certain developed nations.
- Supervised Learning, a branch of AI, provides a solution by training algorithms to recognise patterns in historical patient data to predict future health outcomes.

Literature Survey:

S.No.	Author's Name	Publication Year	Description	Limitation
1.	Almansour, A. M., et al.	2024	This study integrates rule-based machine learning (Apriori algorithm) with classification models (SVM, Bootstrap Forest, and Boosted Trees) to identify frequent symptom patterns and predict diseases.	Difficult to predict communicable diseases due to highly overlapping symptoms across different conditions.
2.	Mohamed Sadiq B. and Nahush R. Jain	2023	It evaluates Naive Bayes, Decision Trees, and Random Forests to classify 41 diseases from 132 symptoms, focusing on creating an accessible web interface.	Model performance can be limited by feature dependencies that traditional algorithms like Naive Bayes may fail to capture accurately.
3.	Hardik Sondhi	2025	It introduces a hybrid model using an ensemble of ML algorithms (NB, DT, RF) paired with the DeepSeek Large Language Model (LLM) to provide both a diagnosis and medical guidance.	Accuracy is sensitive to the number of input symptoms; the system is optimised for a limited set (often three) for initial screening.
4.	Priya Mishra, Uday Singh Kushwaha, and Shraddha Singh	2025/26	A comprehensive comparison of Decision Tree (DT), Random Forest (RF), and Naive Bayes (NB). It highlights that RF and DT achieved high accuracy (~95.12%) on 132 binary-encoded symptoms.	The findings are constrained by dataset coverage and the simplicity of binary symptom representation (present vs. absent).
5.	Cheeryal V., et al.	2024	It predicts lifestyle-related diseases using machine learning integrated into a single web application using	An imbalanced class (where some diseases have many more

OBJECTIVE:

- To develop various machine learning models for disease prediction
- To evaluate performance using key metrics
- To develop an improved Ensemble/Hybrid Model that outperforms individual models
- To combine multiple algorithms using stacking/meta-learning for higher accuracy.

Limitations in Existing Models:

- A limited number of samples restricts the model's generalisation to new unseen data
- Rare disease classes can lead to models that perform poorly on these specific conditions.
- High accuracy on a small dataset with many features suggests potential overfitting.
- Complex Ensemble Learning Models made it difficult to understand and predict the diseases.

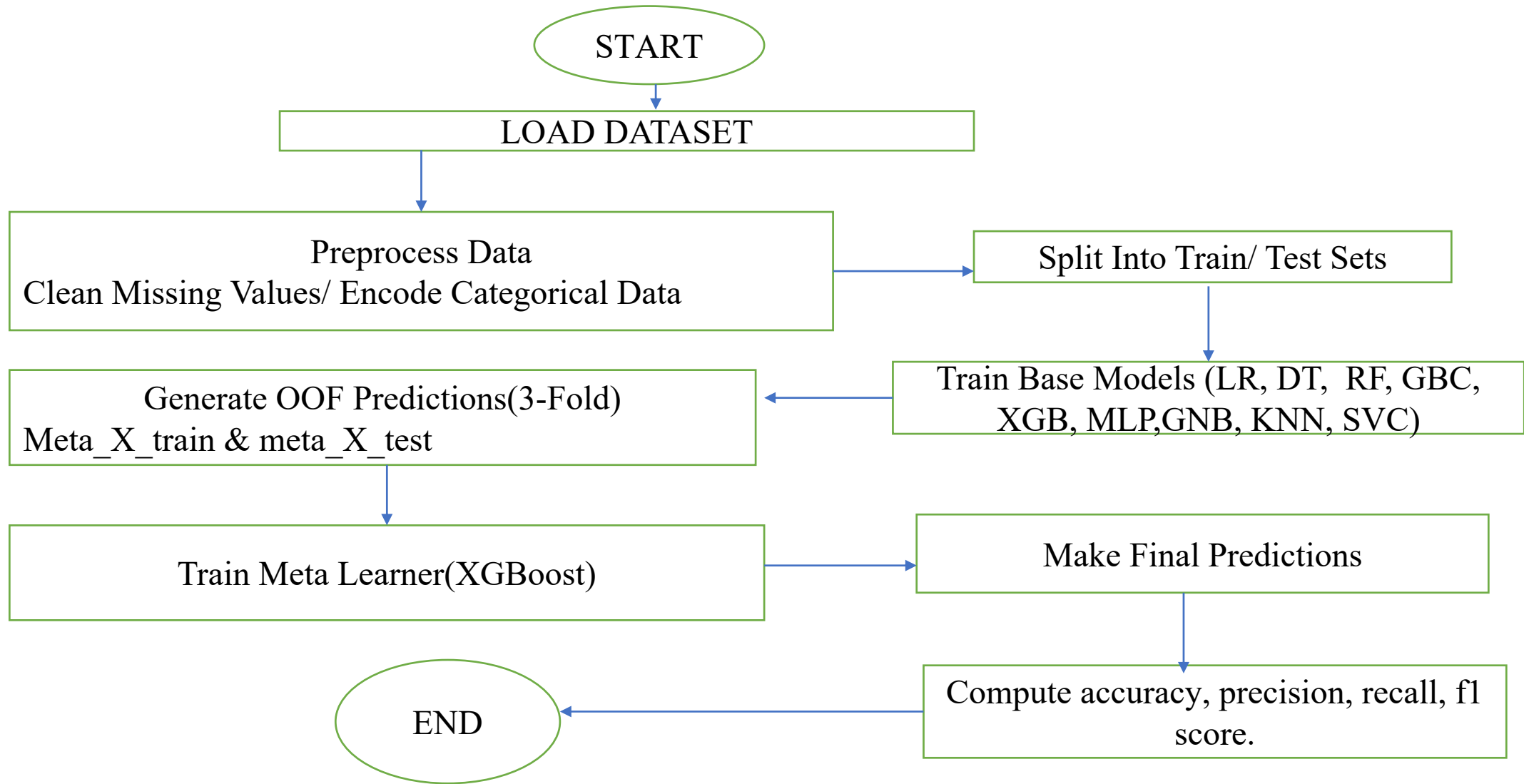
Proposed Models

- 9 Models are used to predict disease: Logistic Regression, Support Vector Machine (SVC), Decision Tree, Random Forest, K-Nearest Neighbors, MLP Classifier, Gradient Boosting, Gaussian Naïve Bayes and XGBoost.
- It achieved an accuracy of 1.0 for Logistic Regression, 0.721 for Decision Tree, 0.967 for Random Forest, 0.885 for Support Vector Machine (SVC), 0.9508 for K-Nearest Neighbors, 1.0 for Gaussian Naïve Bayes, 1.0 for MLP Classifier, 0.8688 for Gradient Boosting and 0.787 for XGBoost.

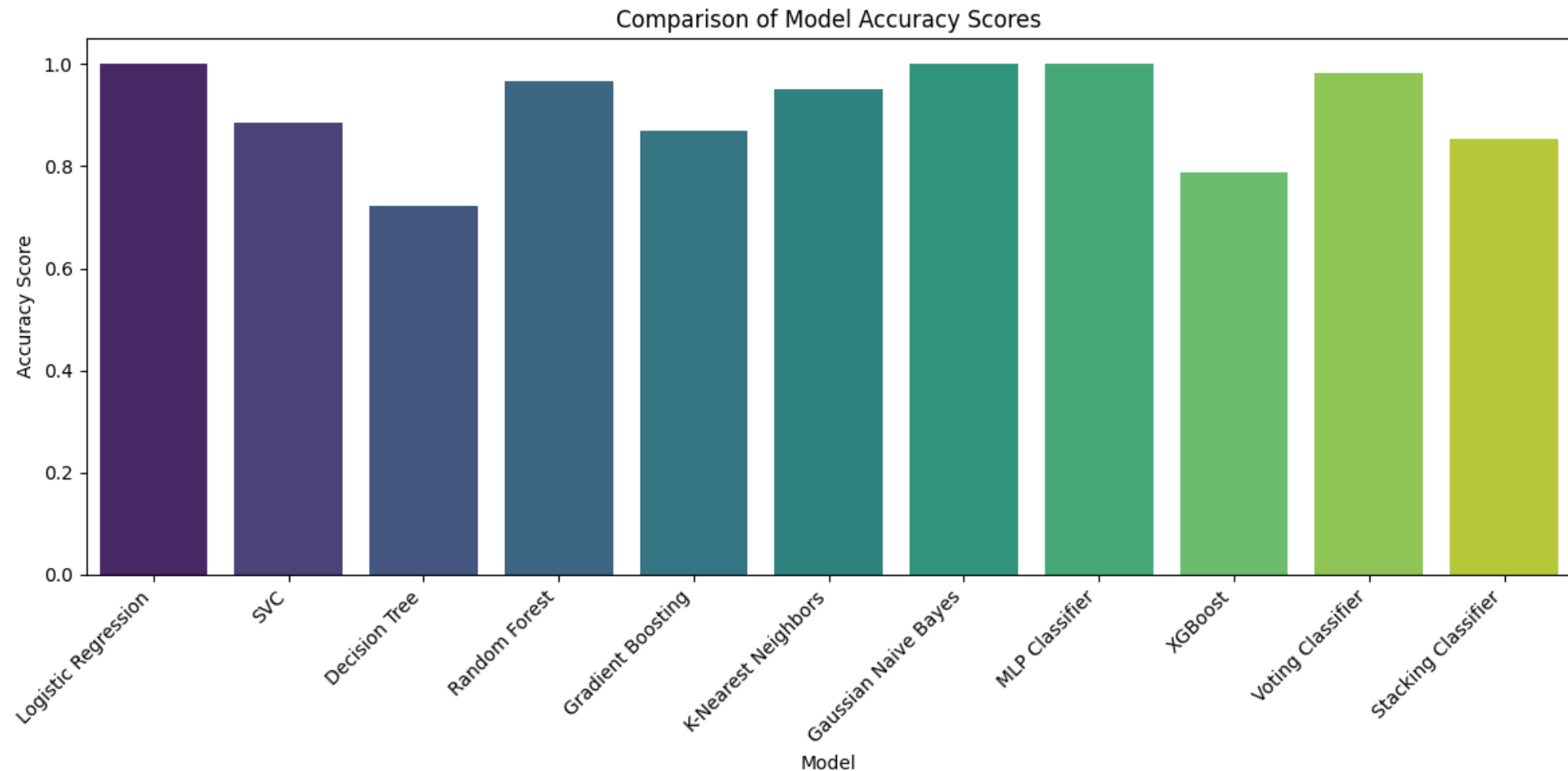
Phases in Proposed Methodology

1. Data Loading: Load the 'DiseaseAndSymptoms.csv' dataset into a pandas DataFrame.
2. Data Preprocessing: Handle missing values, label encode the target variable ('Disease') and one-hot encode categorical variables (Symptoms).
3. Model Training and Evaluation: Train and evaluate a selection of 9 different classification models (e.g., Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, K-Nearest Neighbours, MLP Classifier, Gaussian Naïve Bayes, XGBoost, Gradient Boosting) on the training data, and assess their performance using accuracy on the testing data.
4. Hybrid Meta-Model (Stacking Ensemble): Generate predictions from all 9 base models.
5. Performance Evaluation: Compare the Hybrid Model vs. each base model.
6. Final Disease Prediction: Use the hybrid to predict the disease.

FLOWCHART OF TRAINING PIPELINE



OUTCOME



Conclusion and Future Scope:

- The study shows that ensemble learning algorithms improve disease prediction accuracy compared to individual models.
- Combining multiple base models using stacking and meta-learning results in a more stable and reliable hybrid model.
- Proper data preprocessing, feature engineering, and model evaluation are essential for achieving high performance.
- Synthesise the results from the accuracy comparison and confusion matrices to draw a comprehensive conclusion about the performance of the various models, highlighting the best performers and any surprising outcomes.